PROJECT TITLE:

Royan Décor Sales Performance Analysis: Product Restocking and Payment Method Insights

Subtitle

An end-to-end data analytics project using Excel, SQL, and Power BI to identify top-performing products, evaluate payment methods, and provide actionable recommendations for inventory management and business decision-making.

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PROJECT DEFINITION

EVALUATION OF BUSINESS PROBLEM / QUESTION

<u>What is the key point?</u>

The CEO of Royan Décor wants to restock her store, but she's not sure on which products she should prioritize on. And she has a very limited budget to work with. So, she sent her sales record which was handwritten into a journal in a picture format. This was entered into Excel to create A SALES RECORD. The record contains only records of sales from 14th of April till 28th July, when this analysis was carried out.

She also has an issue with her POS machine and she wants to know if it's worth getting in order, and it depends on usage frequency.

Simply put, she wants to find out;

- Which products should be restocked?
- Should the company keep or remove the POS payment method?

Expansively;

- What is the total sales?
- What is the total revenue generated?
- What are the total sales per month?
- Which products are the top performing products?
- How effective are the transaction types?
- Which product segment (Category) responded best?
- Which product had the best balance between sales and profit?

We will be working on this project with these questions in mind.

OBJECTIVE

Therefore, our objective is:

> Analyze Royan Décor sales data to identify high-performing products and evaluate the effectiveness of different payment methods to support inventory and business decisions.

KEY PERFORMANCE INDICATORS (KPIS)

To understand the how well the business has performed, the products that need to be restocked and the effectiveness of the payment methods, we will be looking at the following KPIs;

- Total sales
- Total Revenue
- Average sales and revenues
- Transaction types

DATA IDENTIFICATION

DATA REQUIREMENTS & SOURCE

What is needed to achieve our objective? This will determine our data source.

Since it is within the company, we will need the company's Sales Data (Sales Record) that includes:

- Total Sales
- Total Profit
- List of products; and
- Transactions and types

The data source is a first party data received from Royan Décor.

AVAILABILTY & COMPLIANCE CHECKS

Availability Check

The required sales data was available for analysis and contained the key fields necessary to evaluate Royan Decor's sales performance. The dataset included information relating to sales dates, products, categories, transaction types, quantities/goods sold, and revenue. These fields provided sufficient information to analyze revenue trends, product performance, category performance, and transaction-type performance.

The available data was also suitable for integration into Power BI for visualization and dashboard development. However, additional financial information such as product cost, operating expenses, and profit margins was not available. Therefore, the analysis focused primarily on revenue and sales performance, while precise profitability and ROI calculations were not established.

Check	Finding
Sales data available	Yes
Product information	Available
Category information	Available
Transaction type	Available
Sales/revenue	Available
Date information	Available
Quantity/goods sold	Available
Cost/profit information	Not available
Data suitable for Power BI	Yes

Compliance Check

This is about checking whether the data follows the required business rules, data standards, and reporting requirements.

The dataset was reviewed against the defined business and data-quality requirements before analysis. Data fields were checked for appropriate formats, consistency, completeness, duplicate records, and valid categorical values. Transaction types were reviewed to ensure that they followed the expected classifications, while dates, quantities, and revenue values were checked for appropriate formats and reasonable values.

The analysis also ensured that the dashboard calculations were based on the validated dataset and that the reporting period was consistent with the available sales records. Any data-quality issues identified during validation were addressed before the data was used for the final analysis.

Compliance Check	What was checked
Data types	Dates, numbers and text stored correctly
Missing values	Checked important fields for blanks
Duplicates	Checked for duplicate transactions
Dates	Checked that dates were valid and within the analysis period
Revenue	Checked that revenue values were numeric and reasonable
Quantity	Checked that goods sold values were valid
Transaction type	Checked TRF, POS, CASH and TRF & CASH classifications
Categories	Checked that products were assigned to appropriate categories
Consistency	Checked naming and formatting consistency
Calculations	Cross-checked dashboard measures against source data

DATA EXPLORATION

Data Exploration Notes

- The data contains 76 rows and 8 columns.
- There were 35 blank columns in the “Description” Column which is to be left blank since there’s no description available for it.
- There’s no duplicate transaction(s) found in the dataset (since it was manually recorded).
- The “description” is not necessary and would be removed.
- There are some columns that are not in the right format, they’ll need to be addressed.
- The transaction types include the following types of transactions;
    - TRF – transactions sent via electronic bank transfer
    - POS – transactions performed using a debit card
    - CASH – transactions completed with physical cash

DATA ACQUISITION & PREPARATION

COLLECTION & EXTRACTION

Tools Used

Microsoft Excel was used to;

- Record the data
- Explore the data
- Format some certain columns in order to properly import the dataset into SQL.

MySQL Server was used to;

- Clean, test and analyze the data

Power BI was used to visualize the data via interactive dashboards

Datascienceportfol.io was used to host the project.

DATA CLEANING

- We tried to import the data into SQL workbench but we ran into an issue so we had to format some certain data in the “Purchase Amount” column. The data was converted to CSV file and was successfully imported into MySQL workbench.
- Changed the data type of the columns in our database to the right formats.

```
alter table royandecorsales
modify column category varchar(100),
modify column Quantity decimal (10, 2),
modify column Measure varchar(100),
modify column purchaseamount int,
modify column date date,
modify column transactiontype varchar(100)
```

- The “description” and “measure” columns as they won’t be needed for this analysis. And this was done by pulling up the columns we needed in the database;

```
select
   goods,
   quantity,
   Purchaseamount,
   date,
   transactiontype
 from
   royandecorsales
```

- Checked for blanks in the work book;

```
select *
From royandecorsales
where goods is null
  or quantity is null
  or purchaseamount is null
```

- Removed any space within the goods column that might affect our analysis.

```
update royandecorsales
set goods = trim(goods);
```

- Aff

CREATE A VIEW TO STORE THE TRANSFORMED

```
/*
# 1. Create a view to store the transformed data
# 2. Select the required columns from the royandecorsales SQL table
*/

-- 1.
CREATE VIEW view_royandecorsales AS
-- 2.
SELECT
    goods,
    Category,
    Quantity,
    PurchaseAmount,
    Date,
    TransactionType


FROM
    Royandecorsales
```

DATA VALIDATION & QUALITY CHECK

What tests do we need to validate our data sets?

To continue this analysis, we need to be sure that our data meets the requirements to generate the insights we need.

We will be carrying out the following data tests;

1. Column count check
2. Row count check
3. Data type check
4. Duplicate check

COLUMN COUNT CHECK

SQL Query

```
/*
# Count the total number of columns (or fields) in the SQL view
*/
SELECT
   COUNT(*) AS column_count
FROM
   INFORMATION_SCHEMA.COLUMNS
WHERE
   TABLE_NAME = 'view_royandecorsales'
```

Output

ROW COUNT CHECK

SQL Query

```
/*
```

```
# Count the total number of records (or rows) in the SQL view
*/
SELECT
   COUNT(*) AS no_of_rows
FROM
   view_royandecorsales;
```

Output

DATA TYPE CHECK

SQL Query

```
/*
# Check the data types of each column from the view by checking the INFORMATION SCHEMA view
*/-- 1.

SELECT
   COLUMN_NAME,
   DATA_TYPE
FROM
   INFORMATION_SCHEMA.COLUMNS
WHERE
   TABLE_NAME = 'view_royandecorsales';
```

Output

DUPLICATE CHECK

SQL Query

Output

VISUALIZATION

RESULTS

DAX MEASURES

1. Total Revenue
2. Total Quantity Sold
3. Total Transactions
4. Average Transaction Value
5. Percentage Contribution of each Transaction type
6. Average Quantity per Transanction

1. Total Revenue

Total Revenue =

VAR Totalrevenue = SUM('royandecor view_royandecorsales'[PurchaseAmount])

RETURN Totalrevenue

---

2. Total Quantity Sold

Total Quantity Sold =

VAR Totalquantitysold = SUM('royandecor view_royandecorsales'[Quantity])

RETURN Totalquantitysold

Shows how many items were sold.

---

3. Total Transactions

Total Transactions =

VAR TotalTransactions = COUNT('royandecor view_royandecorsales'[TransactionType])

RETURN TotalTransactions

Shows the number of sales transactions.

---

4. Average Transaction Value

Average Transaction Value =

VAR TotalRevenue = SUM('royandecor view_royandecorsales'[PurchaseAmount])

VAR TotalTransaction = SUM('royandecor view_royandecorsales'[TransactionType])

VAR AvgTransVal = DIVIDE([Total Revenue], [Total Transactions])

RETURN AvgTransVal

This tells the CEO how much the average customer spends per purchase.

---

5. Percentage Contribution of Each Transaction Type

Percentage Contribution of Each Transaction Type =

VAR TotalRevenue = SUM('royandecor view_royandecorsales'[PurchaseAmount])

VAR PercentContributionofTransaction =

```
DIVIDE(
   TotalRevenue,
   CALCULATE(
      TotalRevenue,
      ALL('royandecor view_royandecorsales'[TransactionType])
))

RETURN PercentContributionofTransaction
```

---

6. Average Quantity per Transaction

```
Average Quantity =
AVERAGE('royandecor view_royandecorsales'[Quantity])
RETURN AverageQuantityPerTransaction
```

ANALYSIS

EXPLORATORY DATA ANALYSIS

Findings

For this analysis, we will be focusing on the questions that it should be answered. Which include;

- What is the total sales?
- What is the total revenue generated?
- What are the total sales per month?
- Which products are the top performing products?
- How effective are the transaction types?
- Which product segment (Category) responded best?

- Which product had the best balance between sales and profit?

TOTAL SALES

TOTAL SALES PER MONTH

TOP PERFORMING PRODUCTS

TRANSACTION TYPES (Effectiveness)

PRODUCT CATEGORIES

SALES TO REVENUE RATIOS

DISCOVERY

The analysis of Royan Decor's sales performance reveals several important patterns:

- **Total revenue:** The business generated approximately **₦1.17 million** from **142 sales** across **75 transactions**, with an average transaction value of approximately **₦15.61K**.
- **Category performance: Flooring** generated the highest revenue, followed by **Home Accessories**, **Flower & Accessories**, and **Wallpapers**. This indicates that flooring is currently a major revenue contributor.
- **Product performance: Carpet Grass** is the highest-revenue product, followed by **Centre Rug**, **Marble Sticker**, and **Wallpaper**. These products appear to be important contributors to overall revenue.
- **Transaction type:** The **TRF** transaction type accounts for approximately **₦899K (76.8%)** of revenue, considerably higher than POS and CASH transactions. This indicates a strong concentration of revenue in this transaction channel. However, with **POS** transactions generating **₦143K (12.22%)**, it is worth paying attention to.
- **Sales trend:** Monthly revenue fluctuates considerably, with several periods showing sharp increases followed by declines. This suggests that sales performance is not consistent across the analyzed period.
- **Product concentration:** A relatively small group of products contributes a substantial portion of revenue. This creates an opportunity to focus inventory and promotional activities on high-performing products while monitoring lower-performing products.

The analysis indicates that Royan Decor's revenue is concentrated around a small number of product categories and products, particularly Flooring and Carpet Grass. In addition, the POS transaction type contributes approximately 12.22% of total revenue. However, monthly sales show considerable fluctuations, suggesting an opportunity to improve sales consistency through targeted inventory management, product promotion, and monitoring of transaction channels.

RECOMMENDATIONS

I) Potential Courses of Action

Main Potential Courses of Action

1. Strengthen the Flooring category

Since Flooring is the highest-revenue category, management could allocate more inventory and promotional resources to this category.

Potential actions:

* Maintain sufficient flooring inventory.
* Introduce complementary products.
* Offer bundle pricing where appropriate.
* Monitor flooring sales monthly to identify demand patterns.

2. Desist from the removal of POS transaction type (Method)

Since POS accounts for around 12.22%, it might be worth keeping other than removing it. Since it has proven to be useful in the payment process.

The objective then, is to:

* Work on improving the functionality of the POS payment method.
* Understand why TRF is used more often.
* Find out whether the transaction classification is being recorded consistently.

This could improve both financial reporting and transaction management.

Other Possible Potential Courses of Action

1. Increase focus on high-performing products

Royan Decor should prioritize products such as Carpet Grass, Centre Rug, Marble Sticker, and Wallpaper, which contribute significantly to revenue.

Potential actions:

* Maintain adequate stock levels for high-demand products.
* Identify products frequently purchased together.

- Create bundles around high-performing products.
- Promote high-performing products through targeted marketing.

2. Improve sales consistency

The monthly sales chart shows substantial fluctuations.

Management could:

- Identify the reasons behind monthly peaks.
- Determine whether peaks are caused by specific products, customers, promotions, or large transactions.
- Replicate successful sales strategies during weaker periods.
- Develop monthly sales targets.

3. Review low-performing products

Products with consistently low revenue should be reviewed before additional inventory is purchased.

Management could classify products into:

- High-performing → Maintain/increase stock
- Moderate-performing → Monitor/promote
- Low-performing → Review, discount, bundle, or reduce restocking

II) Estimated ROI & Business Impact

Estimated ROI

The standard formula is:

$$ROI = (Return - Investment) \div Investment \times 100$$

The Sales Record received doesn't contain the amount invested, profit generated. So we can't get a proper ROI. So, we will be developing some scenarios to use and measure potential ROI.

Business Scenario

The proposed actions are expected to improve revenue through better inventory allocation, increased availability of high-performing products, and targeted promotion. For example, if these initiatives generate a conservative 10% increase in revenue on the current ₦1.17M

revenue base, the potential additional revenue would be approximately ₦117,000. However, the actual ROI cannot be determined without information on implementation costs, product costs, gross margins, and operating expenses.

Potential revenue:

Current revenue = ₦1.17M

10% potential increase = ₦117K

Potential revenue after increase = ₦1.287M

Other Business scenarios:

Scenario	Revenue Increase	Potential Additional Revenue
Conservative	5%	~₦58,500
Moderate	10%	~₦117,000
Higher-impact scenario	15%	~₦175,500

These are scenario estimates, not actual forecasts.

Business impact

The expected business impact could include:

* Improved inventory allocation (proper product stocking).
* Efficient usage and management of the various payment methods, particularly the POS payment method.
* Reduced risk of stockouts for high-performing products.
* Better identification of products requiring promotional attention.
* More consistent monthly sales.
* Improved understanding of transaction channels.
* Better sales and inventory planning.
* Potential increase in revenue and profitability.

CONCLUSION

The analysis of Royan Decor's sales performance shows that the business generated approximately ₦1.17 million in revenue from 142 sales and 75 transactions. Revenue is concentrated primarily in the Flooring category, with products such as Carpet Grass and Centre Rug contributing significantly to overall sales. The TRF transaction type also represents the largest proportion of revenue at approximately 76.8%. However, the POS payment method has proven useful and shouldn’t be removed, rather let it be efficiently managed for continuity and diversification.

The analysis further indicates considerable variation in monthly sales performance, creating an opportunity for the business to improve sales consistency. Management should prioritize high-performing products and categories, maintain appropriate inventory levels, investigate the reasons behind the dominance of TAF transactions, and develop strategies for improving weaker-performing products and sales periods.

If effectively implemented, these actions could improve inventory utilization, sales consistency, and revenue generation. However, a precise ROI cannot be established from the current dashboard alone because product costs, profit margins, and implementation costs were not included in the analysis.

EXTRAS

1. Design
2. Development
3. Testing
4. Analysis

DESIGN

Dashboard components

What should the dashboard contain based on the requirements provided?

To understand what it should contain, we need to figure out what questions we need the dashboard to answer:

- What is the total sales?
- What are the total sales per month?
- Which products are the top performing products?
- How effective are the transaction types?
- Which product segment (Category) responded best?
- Which product had the best balance between sales and profit?

Dashboard Mockup

- Monthly Sales (Line Chart)
- Sales by Category (Horizontal bar Chart)
- Transaction Types (Pie Chart)
- Top performing products (Horizontal bar Chart)
- Goods By Transaction Type

ISSUES FACED

- One of the major issues faced was working on the sales-to-Ptofit-Ratio. We had this issue because there was no profit column.